

Using metaheuristic algorithms to solve a multi-objective industrial hazardous waste location-routing problem considering incompatible waste types

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ABSTRACT

Rapid progress in technology is a primary cause of acceleration in the rate of the industrial hazardous waste generation all over the world. Management of hazardous waste has magnetized researcher's attention because of its considerable impacts on the economy, ecology, and the environment. In this regard, this paper addresses a new industrial hazardous waste location-routing problem by putting emphasis on some new aspects in its formulation such as considering restriction about the incompatibility between some kinds of wastes and incorporating routing decisions into the model. Simultaneously minimization of three significant criteria, including total cost, total transportation risk of hazardous waste related to population exposure, and site risk persuades authors to implement two multi-objective evolutionary algorithms, Nondominated Sorting Genetic Algorithm (NSGA-II) and Multi-Objective Particle Swarm Optimization (MOPSO) for tackling the problem. The results obtained from experiments on several problem instances confirm the superiority of NSGA-II over MOPSO in terms of most of the evaluation metrics. Therefore, the significance of the paper is firstly the novelty of the model, and secondly, the comparison of two solution methods allows for the identification of the method resulting in the best results.

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1. Introduction

Hazardous Waste Management (HWM) is a global issue that has become a serious concern for industrialized societies during recent years. Wastes could be categorized as hazardous if they possess at least one of the following traits: toxicity, reactivity, ignitability, and corrosiveness. These kinds of wastes are generated by different sources. From large industries to small businesses, households, hospitals, and farms are main sources of hazardous wastes. An inappropriate HWM system as a threat causes substantial harm to the environment, human health, and safety (Herva et al., 2014). Hence, a critical problem for developed countries is managing the hazardous wastes taking account of safety and cost-effectiveness.

Treatment of these wastes and removing corresponding risks are performed mainly using chemical, thermal, and biological

methods (Kim et al., 1990). Ion exchange, precipitation, oxidation and reduction, and neutralization are common chemical methods, and high-temperature incineration is a common thermal method which is used for this purpose. Treatment of certain organic wastes, for instance, those from the petroleum industry is done biologically. (Hu et al., 2013). Because of the widely differing physical and chemical characteristics of hazardous wastes, treatment technologies have to be carefully matched to each waste type, taking into consideration the nature of the wastes, the degree of hazard reduction required, as well as economic, and other factors. A general scheme for the compatibility of treatment technologies as well as recovery processes with different types of hazardous waste has been provided in the literature (Edule, 2001).

HWM as a complex system primarily consists of the collection, transportation, recycling, treatment, and disposal processes. While disposal of waste has become a serious environmental issue and problem continues to increase with development of industries (Cholake et al., 2017), recycling is critically required to minimize pollution problems and treatment costs (Ren et al., 2017). In other

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words, waste utilization would promote a circular economy industry, reduce the consumption of primary energy and natural resources, and reduce global warming (Wiemes et al., 2017). All these components are very closely interlinked and each component can influence the other (Seadon, 2010). Therefore, the quality and safety of HWM are affected by the decisions taken on all of the aforementioned processes (Zhao et al., 2016).

Three main objectives addressed by researchers in the field of HWM are minimizing the overall costs such as constructing and operating costs related to waste management system (Nema and Gupta, 1999), minimizing the immediate or long-term potential risks that are imposed on the surrounding population centers and environment (i Or, 1994), and increasing the equity of the distribution of the mentioned risks imposed by HWM system (Wyman and Kuby, 1995). Therefore, the optimization of the HWM system requires to pay immense attention to the governmental regulation/policy and industrial/public consciousness (Tsai and Chou, 2004).

The studies associated with HWM optimization problems are classified into three main categories. Because of the importance of optimization of locating decisions in HWM systems, the first category of these studies are related to the location planning (Sumathi et al., 2008). For example, Jabbarzadeh et al. (2016) developed a multi-objective model in order to determine suitable locations for waste processing facilities. Minimization of the total costs, greenhouse gas emissions, and fuel consumption are the significant criteria considered in this study. The second category of HWM studies concentrates on the routing planning. Minimizing the total traveling time and number of applied vehicles considering characteristics of wastes was examined by Minh et al. (2013). Decreasing the negative impacts on the surrounding population centers and environment is the main criterion in a waste collection problem (Ristić et al., 2016). Integration of the two mentioned HWM problems called Hazardous Waste Location-Routing Problems (HWLRPs) and classified as the last category. In comparison to considering location and routing problems separately, modeling the two problems simultaneously could lead to more efficient solutions. In other words, each problem contains complex decisions and the decisions on each problem affect another problem.

The first effort to address a HWLRP with a single type of hazardous waste is related to Zografos and Samara (1989). To locate the disposal facilities and to determine the efficient routes between demanding nodes and disposal facilities, a goal-programming model was proposed by authors which optimizes the travel time, transportation risk, and site risk. Alumur and Kara (2007) addressed a multi-objective location-routing problem to manage multiple types of hazardous wastes. In order to design an effective HWM system, they incorporated some constraints which have been ignored in the

literature into their model. For example, the type of technology for each treatment center was considered as a decision variable and the compatibility between waste and treatment technology was addressed as a linear constraint in their proposed model. The problem of establishing incinerators by considering standards of the air pollution was addressed by Emek and Kara (2007) in a HWLRP incorporating recycling and treatment facilities. To achieve the minimum transportation cost and to satisfy the constraint of air pollution, they developed an integer programming model. The work of Zhang and Zhao (2011) is an example for HWLRP under the uncertain conditions. Minimizing the total costs and maximizing risk equality were the objectives of their study. A comprehensive overall of various objective functions including operation costs for different HWM system facilities, revenues from selling recycled wastes, initial investment costs for opening each center distinguishes the work of Boyer et al. (2013) from the previous studies in this area. Considering direct routes for shipping wastes to disposal centers was another feature of their work. In order to complete the HWM network, Samanlioglu (2013) considered direct routes from recycling centers to disposal centers for shipping non-recyclable wastes. Assessing site risk for different facilities is another contribution of that work. That work was extended by introducing fuzzy satisfaction concept and human feeling factors to the model by Ghezavati and Morakabatchian (2015). In addition to generation nodes, several nodes called warehouse nodes were considered in the network which wastes are gathered. In some previous studies, the routes are supposed to be a tour. For example, Zhao and Verter (2015) examined a used oil location-routing problem with the same starting and ending points for vehicles. Significant features of the aforementioned studies on a HWLRP and the current study are shown in Table 1.

As can be founded from Table 1, incompatible characteristics of different types of hazardous wastes have not been considered in the previous HWLRPs. Furthermore, these studies did not incorporate the routing decisions into the designing phase of the HWM system. With respect to these gaps as well as some realistic assumptions disregarded in the previous studies, some distinguishing features of this study are as follows:

- addressing different types of wastes
- considering waste-waste compatibility requirements
- incorporating the collection vehicle routing decisions into the problem
- considering a heterogeneous fleet of vehicles in terms of capacity and maximum allowable travelled distance
- application of multi-objective evolutionary algorithms to handle the HWLRPs

Table 1
Significant feature of this study and relevant hazardous waste location-routing problems.

Study	System framework			Compatibility		Route			Vehicle		Objective		
	Treatment	Recycling	Disposal	Waste- Waste	Waste- Technology	Generation- Facility	Facility- Facility	Generation- Generation	Homogeneous	Heterogeneous	Cost	Risk	Equity
Zografos and Samara (1989)			✓		✓	✓			✓		✓	✓	
Alumur and Kara (2007)	✓		✓			✓	✓		✓		✓	✓	
Emek and Kara (2007)	✓	✓	✓	✓		✓	✓		✓		✓	✓	✓
Zhang and Zhao (2011)	✓		✓	✓		✓	✓		✓		✓	✓	
Boyer et al. (2013)	✓	✓	✓	✓		✓	✓		✓		✓	✓	
Samanlioglu (2013)	✓	✓	✓	✓		✓	✓		✓		✓	✓	
Ghezavati and Morakabatchian (2015)	✓	✓	✓			✓	✓	✓	✓		✓	✓	
Zhao and Verter (2015)	✓		✓	✓		✓	✓		✓		✓	✓	
This study	✓	✓	✓	✓	✓	✓	✓	✓		✓	✓	✓	✓

The remainder of this paper is structured as follows: Problem description and mathematical formulation are provided in Section 2. The solution approach for the developed model is presented in Section 3. Section 4 is assigned to the parameters tuning for NSGA-II and MOPSO algorithms. Model validation, numerical results for several problem instances, and multi-objective evolutionary algorithms comparison are presented in Section 5. Finally, Section 6 concludes the paper and presents directions for future research.

2. Problem description

The problem considered in this paper tries to develop a new formulation of location-routing problem in the case of HWM system. The rise of concern about the collection of commingled incompatible hazardous waste types and its potential risks motivate us to incorporate some new restrictions into the problem.

The model can manage different types of incompatible wastes and handle waste-waste compatibility considerations. In this regard, to avoid the interaction between different wastes such as gas emission and heat generation, a heterogeneous fleet of collection vehicles is utilized in order to gather waste in separate vehicles which are compatible with their loads. The vehicles associated with each type of waste have specific limitations on their capacity and tour length. HWM system in this problem is composed of five main groups of components, including a central depot, a set of demanding nodes, and three sets of facilities (treatment, recycling, and disposal centers). Note that each set of facilities includes the potential facilities can be chosen to establish in the future and the operationalized facilities. In our location-routing problem, each collection route starts and finishes at the central depot. After the loading accumulated wastes at a set of demanding nodes by a vehicle, there are two options for dropping off its loads based on waste type: (1) move to a treatment facility with compatible technology with its loads and (2) move to a recycling center. At the treatment centers, treatment process leads to produce two kinds of waste residues including recyclable and non-recyclable. Recyclable waste residues are shipped to recycling centers and other are transported to disposal centers. At recycling centers, waste residues produced during recycling process are also shipped to disposal centers.

While seeking to achieve economic goals of the system, we need to consider that there are many risks and we have challenges to develop decisions to reduce the risks (Wu et al., 2013). Therefore, two types of risks; that is, transportation risk and site risk are taken into account in the problem. The transportation risk is considered for all routes between generation nodes, treatment centers, and disposal centers. To quantify the transportation risk, the concept of population exposure risk is used. This type of risk considers the total number of people along a route during transporting hazardous waste. The site risk for each treatment and disposal facility is measured through the product of the amount of hazardous waste or waste residues available at each facility and the number of people within a certain distance of it. Other assumptions of this study are summarized as follows:

- Four types of wastes are considered:
 1. recyclable waste
 2. non-recyclable waste that is compatible with incineration technology
 3. non-recyclable waste that is compatible with chemical technology
 4. non-recyclable waste that is compatible with both technologies.
- All of the waste types are generated at each demanding node.

- The total amount of each waste type generated at source nodes does not exceed the capacity of vehicles devoted for collecting that type of waste.
- For each type of waste, each demanding node is served exactly once by only one vehicle.
- The partial waste collection is not permitted.
- Vehicles associated with each waste type are homogeneous in terms of capacity and maximum allowable travelled distance.
- Facilities in the network including treatment, recycling, and disposal facilities have limited capacities.
- The number of potential facilities which can be established is limited.
- There exist two kinds of treatment technologies, and at most one of them could be established at each treatment center.
- All of the parameters assumed to be deterministic.
- Transportation costs are estimated by traveling distance

2.1. Mathematical model

By considering the aforementioned assumptions, we now propose a new multi-objective Mixed Integer Programming (MIP) model for the industrial HWLRP that extends the formulation proposed by Samanlioglu (2013) to plan about the following issues efficiently:

- Determining the routes of collection vehicles to collect each type of hazardous waste from generation nodes
- Locating the HWM system facilities including treatment, recycling, and disposal centers
- Determining the technologies adopted at treatment centers
- Routing the waste residues generated at different facilities

Formulation of a mathematical programming model for this system is done based on the notations given in Table 2. The objectives and constraints of this model are as Equations (1a)–(35).

$$\min f_1(x) = \sum_{i \in G} \sum_{j \in G \cup T \cup R} \sum_{k \in K} c_{ij} x_{ijk} l_{ik} \quad (1a)$$

$$+ \sum_{i \in T} \sum_{j \in R} c_{ij} x_{trij} \quad (1b)$$

$$+ \sum_{i \in T} \sum_{j \in F} c_{ij} x_{tfij} \quad (1c)$$

$$+ \sum_{i \in R} \sum_{j \in F} c_{ij} x_{rfij} \quad (1d)$$

$$+ \sum_{q \in Q} \sum_{i \in T-T'} f_{tqi} t_{qi} \quad (1e)$$

$$+ \sum_{i \in R-R'} f_{ri} r_i \quad (1f)$$

$$+ \sum_{i \in F-F'} f_{fi} f_i \quad (1g)$$

$$\min f_2(x) = \sum_{i \in T} \sum_{w \in W} PA_i x_{twi} \quad (2a)$$

$$+ \sum_{i \in F} PA_i x_{fi} \quad (2b)$$

Table 2
Problem's notations.

Notation	Definition
Sets	
$G = \{1, \dots, g\}$	source nodes
$T = \{1, \dots, t\}$	total treatment nodes
$T' \subset T$	operationalized treatment nodes
$R = \{1, \dots, r\}$	total recycling nodes
$R' \subset R$	operationalized recycling nodes
$F = \{1, \dots, f\}$	total disposal nodes
$F' \subset F$	operationalized disposal nodes
$Q = \{1, \dots, q\}$	treatment technologies
$D = \{1, \dots, d\}$	depots
$W = \{1, \dots, w\}$	waste types
$K = \{1, \dots, k\}$	fleet of collection vehicles
Input Parameters	
c_{ij}	transportation cost of one unit of waste from node i to node j
dis_{ij}	travel distance from node i to node j
ft_{qi}	establishment cost of a treatment facility with technology $q \in Q$ at node $i \in (T - T')$
fr_i	establishment cost of a recycling facility at node $i \in (R - R')$
ff_i	establishment cost of a disposal facility at node $i \in (F - F')$
pop_{ij}	population within a given distance of route $i - j$
PA_i	population around node $i \in (T \cup F)$
d_{wi}	amount of waste type $w \in W$ accumulated at generation node $i \in G$
β_{wq}	proportion of recycling for waste type $w \in W$ treated with technology $q \in Q$
r_{wq}	proportion of mass reduction for waste type $w \in W$ treated with technology $q \in Q$
w_i	proportion of total waste recycled at node $i \in R$
tc_i	capacity of treatment facility at node $i \in T$
tc_i^m	minimum capacity required to establish a treatment facility at node $i \in T$
rc_i	capacity of recycling facility at node $i \in R$
rc_i^m	minimum capacity required to establish a recycling facility at node $i \in R$
fc_i	capacity of disposal facility at node $i \in F$
fc_i^m	minimum capacity required to establish a disposal facility at node $i \in F$
com_{wq}	1 if waste type $w \in W$ is compatible with treatment technology $q \in Q$; 0 otherwise
v_{wk}	1 if waste type $w \in W$ is compatible with vehicle $k \in K$; 0 otherwise
a_{qi}	1 if treatment technology $q \in Q$ is available at operationalized treatment facility $i \in T'$; 0 otherwise
δ_w	maximum capacity of a vehicle is compatible with waste type $w \in W$
γ_w	maximum allowable traveling distance of a vehicle is compatible with waste type $w \in W$
Decision variables	
x_{ijk}	1 if node j visited just after node i by vehicle $k \in K$; 0 otherwise
t_{qi}	1 if a treatment facility with technology $q \in Q$ is established at node $i \in (T - T')$; 0 otherwise
r_i	1 if a recycling facility is established at node $i \in R$; 0 otherwise
f_i	1 if a disposal facility is established at node $i \in F$; 0 otherwise
e_{ik}	distance travelled by vehicle $k \in K$ after visiting node i
l_{ik}	load of vehicle $k \in K$ after visiting node i
xt_{wi}	amount of waste type $w \in W$ treated at node $i \in T$
xr_i	amount of waste recycled at node $i \in R$
xf_i	amount of waste disposed at node $i \in F$
xtr_{ij}	amount of recyclable waste residue transported from node $i \in T$ to node $j \in R$
xtf_{ij}	amount of waste residue transported from node $i \in T$ to node $j \in F$
xrf_{ij}	amount of waste residue transported from node $i \in R$ to node $j \in F$

$$\min f_3(x) = \sum_{i \in G} \sum_{j \in G \cup T} \sum_{k \in K} pop_{ij} x_{ijk} l_{ik} \quad (3a) \quad \sum_{i \in G} x_{ijk} - \sum_{i' \in D} x_{ji'k} = 0 \quad \forall j \in R \cup T, k \in K \quad (7)$$

$$+ \sum_{i \in T} \sum_{j \in F} pop_{ij} xtf_{ij} \quad (3b) \quad x_{ijk} \leq \sum_{w \in W} \sum_{q \in Q} v_{wk} com_{wq} t_{qi} \quad \forall i \in G, j \in T, k \in K \quad (8)$$

$$s.t. \quad \sum_{i \in D} \sum_{j \in G} x_{ijk} = 1 \quad \forall k \in K \quad (4) \quad x_{ijk} \leq \sum_{w \in W} v_{wk} \frac{\left(\left(2 - \sum_{q \in Q} com_{wq} \right) r_j \right)}{2} \quad \forall i \in G, j \in R, k \in K \quad (9)$$

$$\sum_{i \in D \cup G} x_{ijk} - \sum_{i' \in G \cup R \cup T} x_{ji'k} = 0 \quad \forall j \in G, k \in K \quad (5)$$

$$\sum_{j \in G \cup R \cup T} \sum_{k \in K} x_{ijk} v_{wk} = 1 \quad \forall i \in G, w \in W \quad (6) \quad e_{ik} - e_{jk} + \sum_{w \in W} v_{wk} \left((\gamma_w + dis_{ij}) x_{ijk} + (\gamma_w - dis_{ij}) x_{jik} \right) \leq \sum_{w \in W} v_{wk} \gamma_w \quad \forall i, j \in G \cup R \cup T \quad (10)$$

$$\sum_{i \in D} \text{dis}_{ijk} x_{ijk} \leq e_{jk} \leq \sum_{w \in W} v_{wk} \left(\gamma_w + \sum_{i \in D} (\text{dis}_{ij} - \gamma_w) x_{ijk} \right) \quad \forall j \in G, k \in K \quad (11)$$

$$e_{ik} \leq \sum_{w \in W} v_{wk} \gamma_w - \sum_{j \in D} \text{dis}_{ij} x_{ijk} \quad \forall j \in R \cup T, k \in K \quad (12)$$

$$l_{ik} - l_{jk} + \sum_{w \in W} v_{wk} \delta_w x_{ijk} \leq \sum_{w \in W} v_{wk} (\delta_w - d_{jw}) \quad \forall i, j \in G, k \in K \quad (13)$$

$$\sum_{w \in W} d_{wi} v_{wk} \leq l_{ik} \leq \sum_{w \in W} v_{wk} \delta_w \quad \forall i \in G, k \in K \quad (14)$$

$$\sum_{i \in D} \sum_{w \in W} x_{ijk} d_{wj} v_{wk} \leq l_{jk} \quad \forall j \in G, k \in K \quad (15)$$

$$l_{jk} \leq \sum_{w \in W} v_{wk} \left(\delta_w + \sum_{i \in D} (d_{wj} - \delta_w) x_{ijk} \right) \quad \forall j \in G, k \in K \quad (16)$$

$$xt_{wj} = \sum_{k \in K} \sum_{i \in G} x_{ijk} l_{ik} v_{wk} \quad \forall j \in T, w \in W \quad (17)$$

$$\sum_{w \in W} xt_{wj} \leq tc_j \quad \sum_{q \in Q} t_{qj} \quad \forall j \in T \quad (18)$$

$$\sum_{w \in W} xt_{wj} \geq tc_j^m \quad \sum_{q \in Q} t_{qj} \quad \forall j \in T \quad (19)$$

$$\sum_{w \in W} \sum_{q \in Q} xt_{wi} t_{qi} (1 - r_{wq}) \beta_{wq} = \sum_{j \in R} xtr_{ij} \quad \forall i \in T \quad (20)$$

$$xr_j = \sum_{w \in W} \sum_{k \in K} \sum_{i \in G} x_{ijk} l_{ik} v_{wk} + \sum_{i' \in T} xtr_{i'j} \quad \forall j \in R \quad (21)$$

$$xr_j \leq rc_j \quad r_j \quad \forall j \in R \quad (22)$$

$$xr_j \geq rc_j^m \quad r_j \quad \forall j \in R \quad (23)$$

$$\sum_{w \in W} \sum_{q \in Q} xt_{wi} t_{qi} (1 - r_{wq}) (1 - \beta_{wq}) = \sum_{j \in F} xtf_{ij} \quad \forall i \in T \quad (24)$$

$$xr_i (1 - w_i) = \sum_{j \in F} xrf_{ij} \quad \forall i \in R \quad (25)$$

$$xf_i = \sum_{j \in T} xtf_{ji} + \sum_{j' \in R} xrf_{j'i} \quad \forall i \in F \quad (26)$$

$$xf_i \leq fc_i \quad f_i \quad \forall i \in F \quad (27)$$

$$xf_i \geq fc_i^m \quad f_i \quad \forall i \in F \quad (28)$$

$$\sum_{w \in W} \sum_{i \in G} d_{wi} = \sum_{w \in W} \sum_{j \in T} xt_{wj} + \sum_{j \in R} xr_j \quad (29)$$

$$\sum_{q \in Q} t_{qi} \leq 1 \quad \forall i \in T \quad (30)$$

$$t_{qi} = a_{qi} \quad \forall q \in Q, i \in T' \quad (31)$$

$$r_i = 1 \quad \forall i \in R' \quad (32)$$

$$f_i = 1 \quad \forall i \in F' \quad xtf_{ij} \geq 0, xrf_{ij} \geq 0 \quad \forall i \in T, j \in F \quad (33)$$

$$xtr_{ij} \geq 0, xt_{wi} \geq 0 \quad \forall i \in T, j \in R, w \in W \quad xrf_{ij} \geq 0, xr_i \geq 0 \quad \forall i \in R, j \in F \quad (34)$$

$$l_{ik} \geq 0 \quad \forall i \in G, k \in K \quad e_{ik} \geq 0 \quad \forall i \in (D \cup G), k \in K \quad x_{ijk} \in \{0, 1\} \quad \forall i \in (D \cup G), j \in (G \cup T \cup R), k \in K$$

$$t_{qi} \in \{0, 1\} \quad \forall q \in Q, i \in T \quad r_i \in \{0, 1\} \quad \forall i \in R \quad f_i \in \{0, 1\} \quad \forall i \in F \quad (35)$$

The first objective function expresses the minimization of the total cost that is formulated with the sum of Equations (1a)–(1g). The cost associated with waste collection phase, including transportation cost of waste between source nodes, the cost of shipment from source nodes to treatment centers with compatible technologies, and the cost of shipment from generation nodes to recycling centers is formulated by Equation (1a). Equations (1b)–(1d) represent the transportation cost of waste residues between different facilities. The fixed cost of establishing system facilities is modeled by Equations (1e)–(1g). The second objective measures the total site risk for the people living around treatment and disposal centers by Equations (2a) and (2b), respectively. Equations (3a)–(3b) form the last objective function and try to minimize the transportation risk for all routes.

Equation (4) states that starting point for all collection vehicles is a central depot. Equation (5) represents one of the traditional VRP constraints guaranteed the continuity of each vehicle's path through arriving and leaving the same generation node. Equation (6) ensures that for each type of waste, all generation nodes are visited exactly once. Equation (7) enforces the vehicles to go back to the depot after unloading the collected wastes at a treatment or recycling center. Equation (8) guarantees that each collection vehicle associated with non-recyclable waste types can only empty its loads at a treatment center with compatible treatment technology. Equation (9) states that the collected recyclable waste types by a vehicle must be delivered to a recycling facility before returning to the central depot. Equations (10)–(12) ensure that the length of routes does not exceed from maximum allowable travelled distance by heterogeneous vehicles fleet (Desrochers and Laporte, 1991). The sets of constraints in Equations (13)–(16) are added to our formulation for sub-tours evasion. Furthermore, these constraints guarantee that a load of collection vehicles does not trespass from their given capacity (Desrochers and Laporte, 1991). The quantity of hazardous waste processed at each treatment center is calculated using Equation (17). The capacity limitations on treatment, recycling, and disposal facilities in terms of minimum and maximum amount of hazardous wastes or waste residues required to establish these facilities are satisfied using Equations ((2), (18), (19), (22) and (23)7–28, respectively. The flow from treatment centers to the recycling facilities is modeled through Equation (20). Equation (21) calculates the amount of recyclable waste type processed at each recycling center. The flow from

treatment and recycling facilities to disposal centers is represented by Equations (24) and (25), respectively. Equation (26) determines the amount of waste residues disposed at each disposal center. Equation (29) ensures that the total demands at different source nodes must be supplied. Among different treatment technologies, at most one of them could be established at a transfer station, this constraint is modeled using Equation (30). By using Equations (31)–(33), existing system facilities are specified. As can be found from these constraints, unlike recycling and disposal facilities, specifying the existing treatment facilities requires to define a binary parameter. Constraints on decision variables are represented by Equations (34) and (35).

The developed MIP model in this sub-section includes nonlinear terms in the first term of objective functions 1 and 3 as well as Equations (17), (20), (21) and (24). By defining some auxiliary variables and constraints we are able to transform the model into a Mixed Integer Linear Programming (MILP) model. Appendix A shows how to linearize the model.

3. Methodology

Location-routing problem belongs to group of NP-hard problems (Yildiz et al., 2012), which means that finding optimal solutions for large-scale problems is generally difficult. Therefore, the NP-hard nature of the problem along with the non-linear nature of the proposed model as well as the existence of multiple contradicting objectives in formulation of the model justify the use of multi-objective evolutionary algorithms in this study. The ability to handle large-scale problems within an acceptable computational time is the main advantage of using these algorithms (Zhou et al., 2011). Hence, two well-known metaheuristic algorithms which have been used extensively by researchers together, i.e. Non-dominated Sorting Genetic Algorithm II (NSGA-II) (Srinivas and Deb, 1994) and Multi-Objective Particle Swarm Optimization (MOPSO) algorithm (Coello et al., 2004) (for example, see Azadeh et al. (2017)) are employed to solve the problem. This section is assigned to a brief description of these algorithms and how to encode individual solutions.

3.1. Solution representation

The performance of the evolutionary algorithms in terms of complexity and computational time depends extremely on the representation scheme (Chen et al., 2013). An appropriate representation scheme in an optimization problem should be capable of representing all the possible candidate solutions. According to the structure of the presented problem, order-based representation is the most adapted method to encode potential solutions of the search space.

In order-based encoding method, each chromosome is defined based on the order of the problem's components. In this study, we consider each chromosome as a $(W + N) \times G$ matrix, in which W , N and G denote the number of waste types, number of different types of system facilities, and number of source nodes, respectively. In this structure, the first W rows of the chromosome are permutations of 1 to G . Next rows of the chromosome are related to the order of system facilities, including treatment, recycling, and disposal facilities. Note that these rows can involve some empty cells. Furthermore, existing facilities should be arranged before potential facilities in the mentioned rows. Fig. 1 shows an illustrative example of a chromosome can be obtained in a solution for a problem with dimension $(12, 4/2, 2/1, 3/2)$ (how to show characteristics of each problem will be explained in Section 5).

Converting this structure into meaningful solutions as well as extracting the decision variables associated with each solution representation need to introduce a decoding procedure. This

2	7	4	3	9	8	10	11	12	5	6	1
order of generation nodes for waste type 1											
5	4	10	12	1	8	6	9	3	2	11	7
order of generation nodes for waste type 2											
6	7	12	10	8	1	5	2	11	9	3	4
order of generation nodes for waste type 3											
3	4	12	2	7	6	10	9	5	8	11	1
order of generation nodes for waste type 4											
3	4	2	1	6	5						
order of treatment facilities											
1	2	3									
order of recycling facilities											
2	1	3	5	4							
order of disposal facilities											

Fig. 1. Solution representation.

procedure involves two stages, including routing stage and transportation stage. To produce feasible route for each collection vehicle in the first stage, we need to determine its destination location. Depending on the waste type, destination is a first-ranked treatment or recycling facility that its left capacity is not less than the capacity of the related collection vehicle. In case of treatment facility, we also search a facility that poses compatible technology with the loads of the collection vehicle. After determining an appropriate facility for unloading the collected wastes, the vehicle's collection route must be determined. For this purpose, we start with the first-ranked unassigned generation node. We continue the assigning procedure until no more nodes can be assigned to the route with respect to the limitations on capacity and maximum distance travelled by collection vehicles.

In the second stage of the decoding procedure, the decision variables associated with the amount of waste residues must be transported between different types of facilities are determined. For this purpose, we start with the first-ranked treatment facility and assign its produced waste residues to the first-ranked disposal center in the disposal centers order. We continue with the next treatment facility of the ordered list and this procedure is iterated until the disposal center does not have enough capacity for the waste residues of the other treatment facilities. By assigning the produced waste residues at all treatment facilities to the disposal centers and applying the same manner to transfer non-recyclable waste residues from recycling facilities to disposal centers as well as recyclable waste residues from treatment facilities to recycling centers this stage is terminated.

3.2. Proposed NSGA-II

The Non-dominated Sorting Genetic Algorithm II (NSGA-II) is a population-based search multi-objective evolutionary algorithm

that has been widely applied in multi-objective optimization problems with two or more conflicting objective functions (Srinivas and Deb, 1994). In such problems, where one global optimum solution that optimizes all objectives simultaneously does not exist, the concept of the Pareto-optimal solution set is applied. A Pareto optimal solution is not dominated by any other solutions and cannot be improved with respect to any objective without deteriorating at least another objective.

A schematic representation of NSGA-II is given in Fig. 2. As you can see in the flowchart, NSGA-II starts with initial population P_0 with population size N_p is generated randomly and the implementation of the non-dominated sorting to categorize the individuals in P_0 into several fronts of non-dominated individuals with respect to the dominance relation in the objective space. Then the crowded-comparison operator is applied to sort the individuals within each front. To obtain the offspring population O_0 with respect to the crossover rate P_c and the mutation rate P_m , firstly the individuals are selected using the tournament selection operator. The main idea behind this selection method is that the individuals

with better rank and less crowded distance has a high probability to select. The next population P_{t+1} is formed from the best individuals of union population $R_t = P_t \cup O_t$ with respect to the non-dominated sorting and crowded distance sorting. The algorithm will be iterated until the stopping criterion will be satisfied. The first rank individuals of the last population are the Pareto optimal solutions.

3.3. NSGA-II operators

To achieve better solutions, the algorithm needs to make sure that the solution space of the problem is explored more completely. For this purpose, NSGA-II uses crossover and mutation operators to create new individuals from old ones. Recombination of the information from a pair of chromosomes during crossover may continue the evolutionary process. In other words, the newborn chromosomes may be having better characteristics compared to their parents. Applying mutation operator guarantees the diversity of solutions as well as preventing the search process falling into a local optimum. These operators and how their implementations will be explained in the following.

3.3.1. Crossover

The implementation of crossover operator and how to produce new offspring from two parents are depicted in Fig. 3. Based on the structure of the chromosomes, single point crossover technique is applied for each row of the chromosome separately. After selecting a crossover point randomly, as you can see in the figure, the first chromosome is copied from beginning to the crossover point and the left information is scanned in the offspring based on their order on the second chromosome. The same manner is applied to produce offspring 2. Note that this approach can be easily implemented on the three last rows of the chromosomes with some empty cells, and therefore applying this crossover technique guarantees the feasibility of the newborn chromosomes.

3.3.2. Mutation

Fig. 4 shows the implementation of mutation operator where two columns of the chromosome are exchanged randomly. If a selected column involves empty cell, the row associated with this cell does not change.

3.4. Proposed MOPSO

As regard to features of Particle Swarm Optimization algorithm (PSO) in solving single objective problems such as simplicity of implementation and coverage of search process to an acceptable solution, Coello et al. (2004) proposed an optimization technique for multiple objective problems, i.e. Multi-Objective Particle Swarm Optimization algorithm (MOPSO). Any solution in this algorithm is a particle that makes a swarm. This optimization method is based on the intelligence group in which each particle has knowledge of its previous best experience (x_{pbest}) and the best global experience of entire swarm (x_{gbest}). Updating the velocity and the position of particle p at iteration t denoted by $v_p(t)$ and $x_p(t)$ is performed based upon Equations (36) and (37).

$$v_p(t) = w \ v_p(t-1) + c_1 \ r_1 \ (x_{pbest}(t) - x_p(t)) + c_2 \ r_2 \ (x_{gbest}(t) - x_p(t)) \quad (36)$$

$$x_p(t) = x_p(t-1) + v_p(t) \quad (37)$$

where the local and global ability of the particle is influenced by the inertia factor w, c_1 and c_2 as the cognitive and social learning

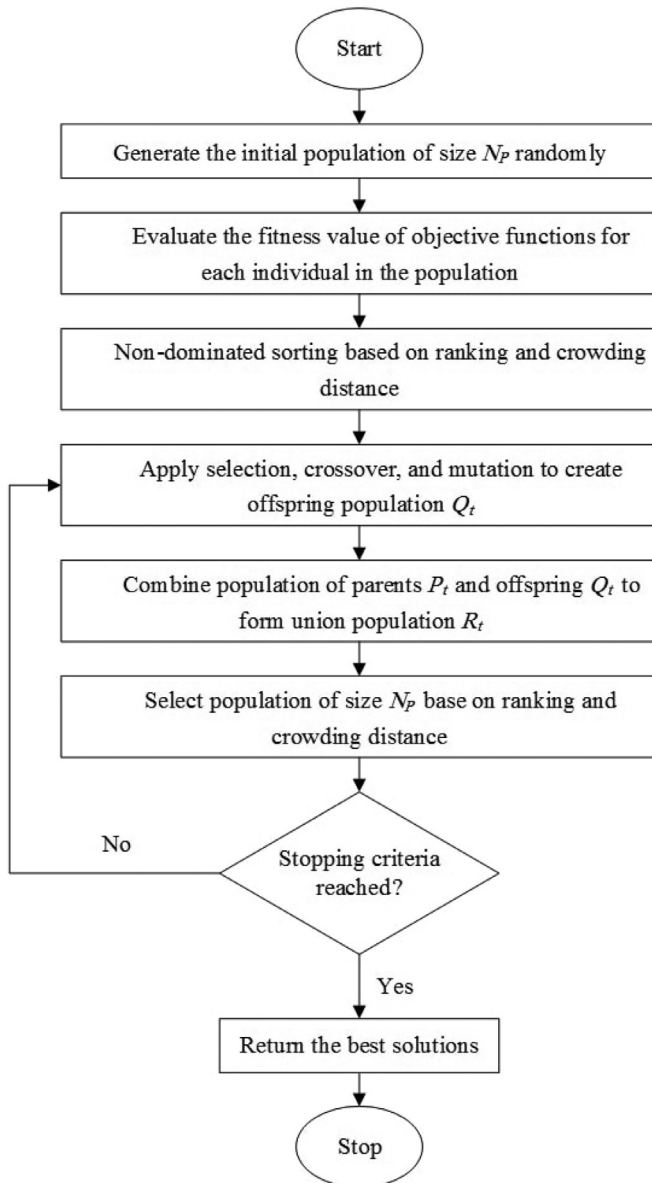


Fig. 2. The flowchart of NSGA-II.

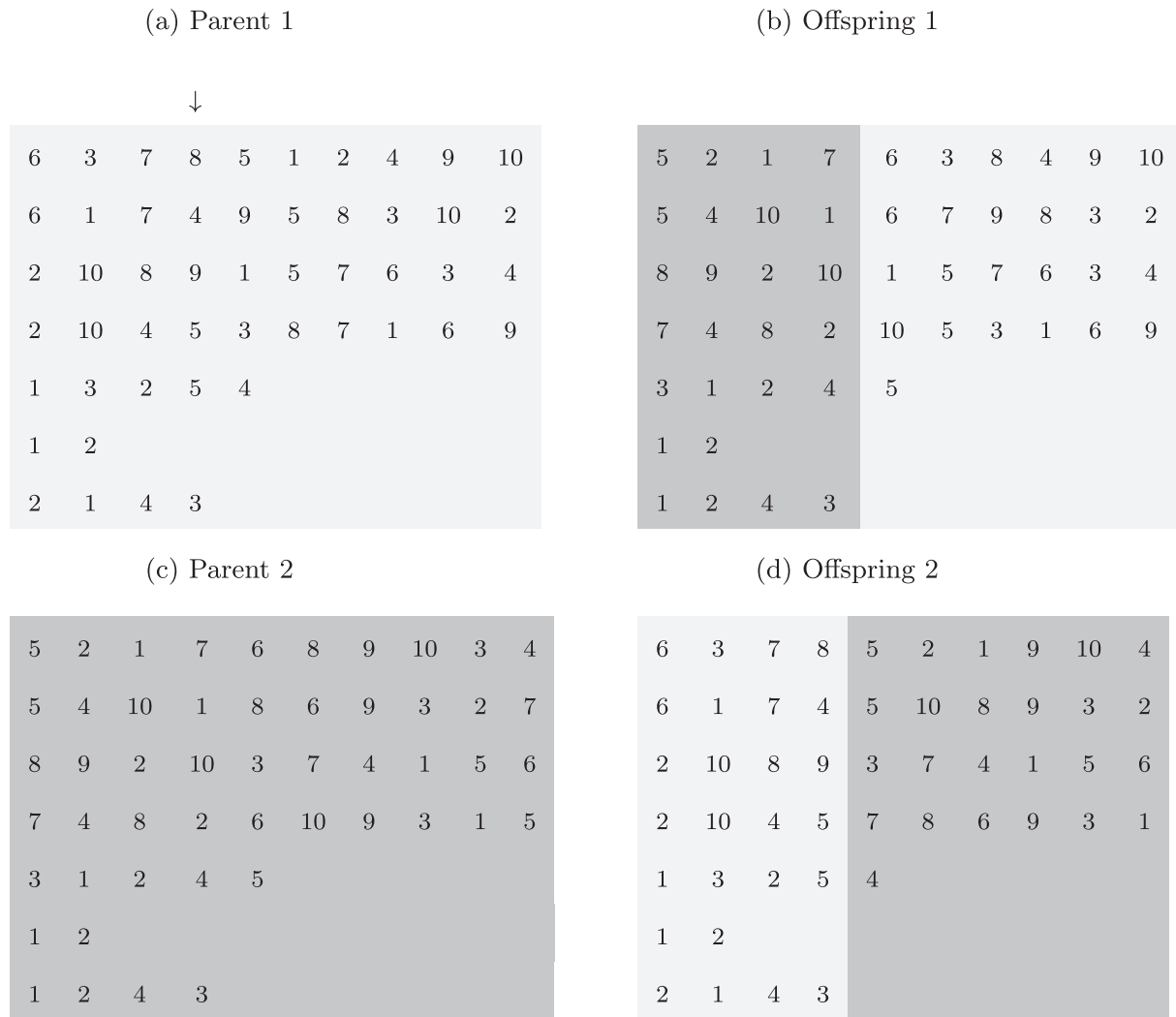


Fig. 3. Single point crossover for order-based encoding method.

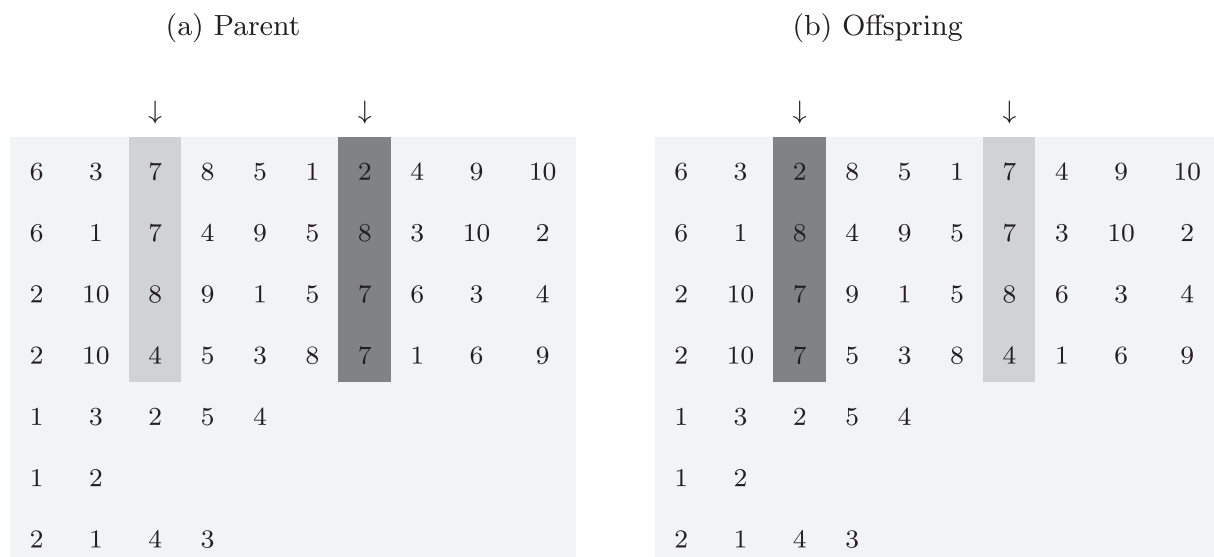


Fig. 4. 2 Single point crossover for order-based encoding method.

coefficients control the effect of x_{pbest} and x_{gbest} respectively. Eventually, r_1 and r_2 are random numbers between 0 and 1. Details about how to implement this algorithm are presented in Fig. 5.

4. Parameters tuning

Since the efficiency and reliability of the evolutionary algorithms are directly influenced by their parameters, calibration of these parameters has got importance (Eiben and Smit, 2011). To improve the experimental results of this study, the Taguchi method (Taguchi, 1986) as a powerful tool for parameter tuning is employed for both algorithms of NSGA-II and MOPSO. The distinctive feature of this method is to achieve the largest information possible with the least number of experiments (Hasçalık and Çaydaş, 2007). In other words, other approaches to statistically design the experimental investigations such as a full factorial experiment become increasingly difficult to carry out investigations when the number of factors becomes considerably large.

A three-level Taguchi design is applied to analyze the effect of essential NSGA-II parameters including, population size (N_p), the total number of iterations (Max_Iteration) as well as crossover and mutation rates (P_c and P_m). The solution quality is considered as the basis of this investigation. In other words, the best value of each objective function is considered as a response for the Taguchi design. The results obtained through implementation of the Taguchi design for medium-scale problem instance 10 in Minitab software are presented in Fig. 6. The same design of experiment for the second solution approach is repeated to determine the optimal levels of swarm size (N_p), the total number of iterations (Max_Iteration), repository size (N_r), inertia weight (w), personal learning coefficient (c_1), and global learning coefficient (c_2). Fig. 7 demonstrates the experimental results of parameters tuning for MOPSO algorithm. Finally, the desirable value of each parameter for both solution approach is reported in Table 3.

5. Results and discussion

5.1. Model validation

To validate the accuracy of the presented model given in Section 2, a small-scale problem (i.e., problem instance 1) is chosen for this purpose and implemented in GAMS software, version 24.1.3. It is necessary to say that this problem is a hypothetical and illustrative example and it may be far away from real industrial case studies based on real-world data. Linearization of the model enables us to solve the problem optimally by the solver CPLEX. With respect to the existence of multiple objective functions in the proposed formulation and the purpose of this Section that is validation of the model, we solve the problem by taking into account only one objective function. In other words, one objective (here, minimization of the site risk) is considered for the problem and then optimized in absence of the other objectives.

Before addressing the problem instances, it is necessary to have some understanding of how to show characteristics of these problems. Each generated instance is characterized with vector $(G, T_1/T_2, R_1/R_2, F_1/F_2)$ that denotes the instance dimension. The first element of this vector denotes the number of generation nodes and the next three elements are related to the treatment, recycling, and disposal facilities, respectively. For these elements, the nominator represents the number of existing facilities and the number of potential facilities is determined by the denominator.

A fleet of collection vehicles including 12 trucks are considered in this instance that covers an area with six demanding nodes (G^1 to G^6). All of waste types can be produced at each node. With respect to the necessity of collecting different waste types separately, each truck is responsible for collecting only one type. In serving the demanding nodes, each vehicle has limitations on capacity and travel distance based on the type of its loads. Also this problem includes operationalized facilities $T_1^1, T_1^2, R_1^1, R_1^2, F_1^1$, and F_2^2 along with some alternatives for establishing new facilities in the future ($T_2^1, T_2^2, R_2^1, R_2^2, F_2^1$, and F_2^2). Full details of the problem are available as Supplementary material.

The optimal solution of the problem in both routing and transportation stages are summarized in Tables 4 and 5. The third column of Table 4 provides the optimal collection routes in which each route starts at a central depot, continues with moving to demanding nodes, and terminates at the same depot. Before returning to the depot, each vehicle empties its loads at a treatment or a recycling facility. Furthermore, the total amount of waste collected by each vehicle as well as its tour length are calculated in the last two columns of this table that verify the feasibility of the obtained solution. The optimal decisions related to the second stage

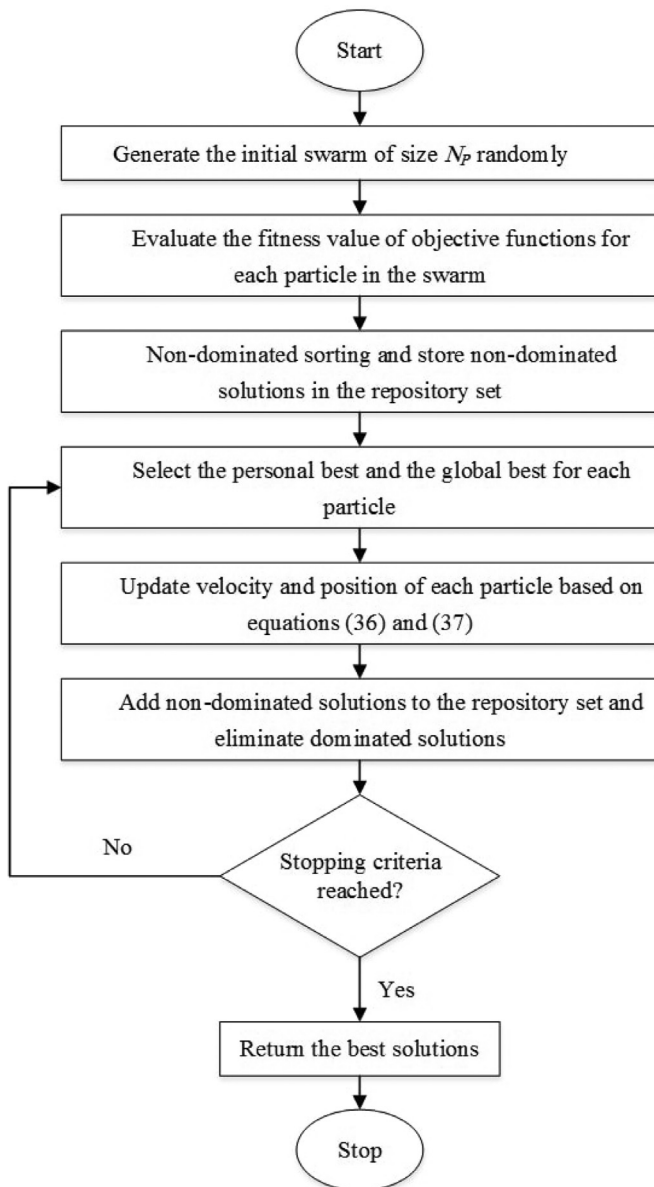


Fig. 5. The flowchart of MOPSO.

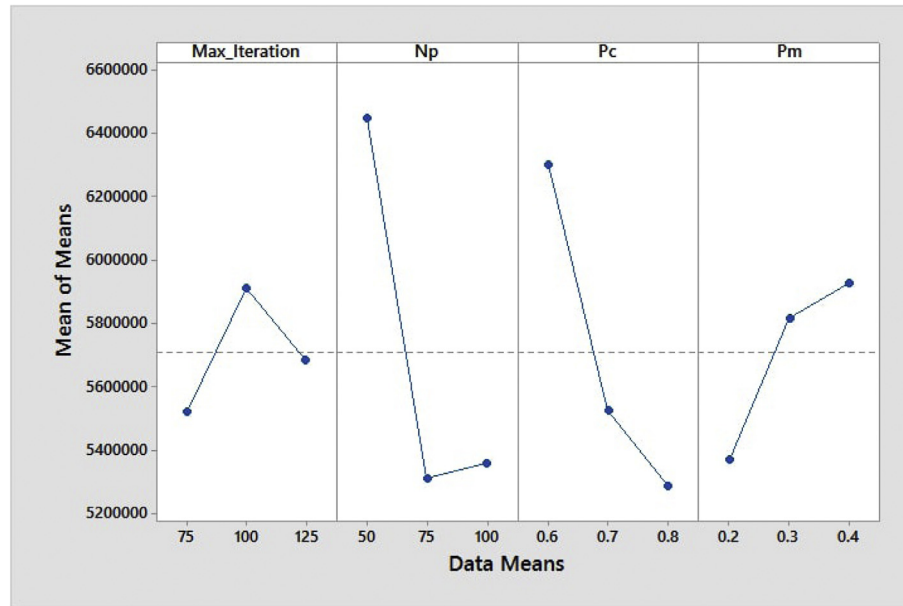


Fig. 6. Analysis diagrams of NSGA-II parameters tuning based on Taguchi method.

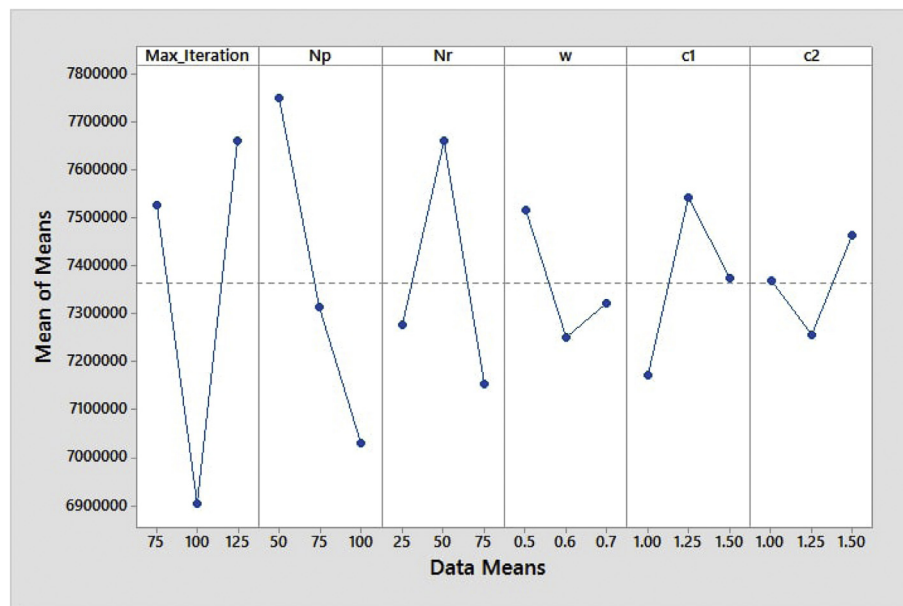


Fig. 7. Analysis diagrams of MOPSO parameters tuning based on Taguchi method.

of the problem including amount of waste processed at each facility and the transportation routes are given in Table 5. This solution leads to a total cost of 4,628,417.888 dollars, global optimum value 11,806,110 $km * people$ for total site risk, and finally a total transportation risk of 87,163,430 $km * people$.

Table 3
Tuned parameters for NSGA-II and MOPSO algorithm.

Algorithm	Parameter							
	Max_iteration	N_p	P_c	P_m	N_r	w	c_1	c_2
NSGA-II	75	75	0.8	0.2	—	—	—	—
MOPSO	100	100	—	—	75	0.6	1	1.25

After validating the proposed model, to assess the effectiveness of the proposed solution approaches, some numerical experiments are provided in this section. Both of the NSGA-II and MOPSO algorithms are coded in MATLAB R2015b software and executed on an Intel Corei7 PC with 8 GB of RAM and over 2 GHz CPU without parallel computing options.

5.2. Test instances

With respect to the lack of benchmark instances in the literature, evaluation of the proposed algorithms is based on three types of problem instances, including small, medium and large scale problems. All these instances are generated randomly using the

Table 4
Optimal decisions for problem instance 1 (routing stage).

Waste type	Vehicle no.	Collection route	Amount of load (100 kg)	Length of route (km)
1	1	depot $\rightarrow G^4 \rightarrow G^3 \rightarrow T_1^1 \rightarrow$ depot	82	44
	2	depot $\rightarrow G^6 \rightarrow G^2 \rightarrow T_1^1 \rightarrow$ depot	61	60
	3	depot $\rightarrow G^1 \rightarrow G^5 \rightarrow T_1^1 \rightarrow$ depot	89	58
2	2	depot $\rightarrow G^3 \rightarrow G^6 \rightarrow T_1^2 \rightarrow$ depot	81	34
	5	depot $\rightarrow G^1 \rightarrow G^5 \rightarrow T_1^2 \rightarrow$ depot	83	51
	6	depot $\rightarrow G^2 \rightarrow G^4 \rightarrow T_1^2 \rightarrow$ depot	92	51
3	7	depot $\rightarrow G^1 \rightarrow G^4 \rightarrow T_1^1 \rightarrow$ depot	89	61
	8	depot $\rightarrow G^3 \rightarrow T_1^1 \rightarrow$ depot	47	37
	9	depot $\rightarrow G^6 \rightarrow G^2 \rightarrow G^5 \rightarrow T_1^1 \rightarrow$ depot	109	61
4	10	depot $\rightarrow G^1 \rightarrow G^5 \rightarrow R_2^1 \rightarrow$ depot	85	51
	11	depot $\rightarrow G^6 \rightarrow G^4 \rightarrow R_2^1 \rightarrow$ depot	81	45
	12	depot $\rightarrow G^3 \rightarrow G^2 \rightarrow R_2^1 \rightarrow$ depot	85	51

Table 5
Optimal decisions for problem instance 1 (transportation stage).

Shipment of waste residues		Processing of wastes	
Transfer route	Amount of load(100 kg)	Facility	Amount of waste processed at each facility(100 kg)
$T_1^1 \rightarrow R_2^1$	95.4	T_1^1	477
$T_1^2 \rightarrow R_2^1$	38.4	T_1^2	256
$T_1^1 \rightarrow F_1^2$	324.360	R_2^1	398.498
$T_1^2 \rightarrow F_1^2$	189.312	F_1^2	553.553
$R_2^1 \rightarrow F_1^2$	59.881		

coded instance generator in MATLAB software and their detailed information are available for download as [Supplementary materials](#).

To be able to evaluate the quality of the solution approaches, each test problem is conducted 10 times independently and the obtained results are averaged. Two samples of non-dominated solution sets for problem instance 10 obtained with the solution approaches are given in Fig. 8.

5.3. Comparison metrics

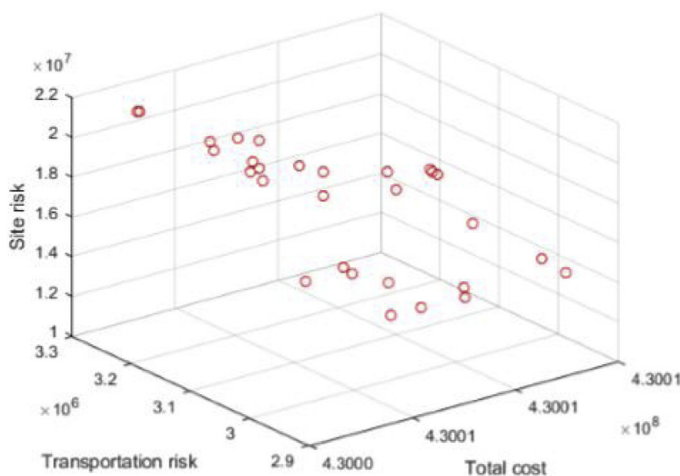
To reveal strengths and weaknesses of the multi-objective evolutionary algorithms, numerous criteria have been presented by researchers. This study employs four comparison metrics to assess the performance of the NSGA-II and MOPSO algorithms in achieving the better solution sets. These metrics are as follows:

- CPU time
- The quantity metric

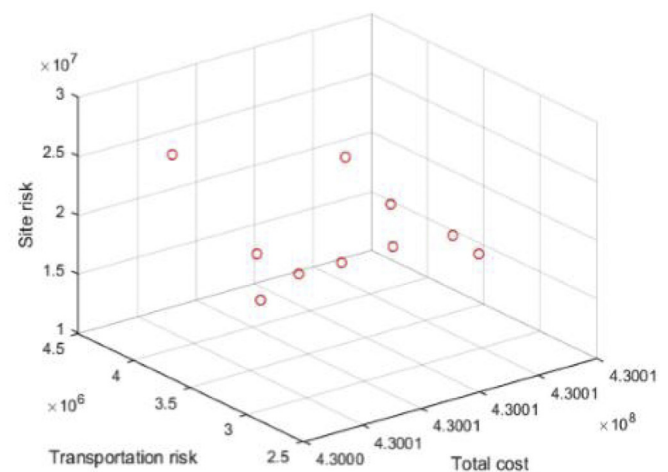
This metric measures the number of Pareto optimal solutions at the simultaneous run for each solution approach. Algorithm with the higher value of this metric can be more converged towards the real Pareto front.

- The spacing metric

Receiving useful information about distribution of non-dominated solutions throughout the obtained Pareto front is done using this metric computed by Equations (38) and (39) (Suo et al., 2017).



(a) NSGA-II



(b) MOPSO

Fig. 8. Approximation of Pareto front for problem instance 10.

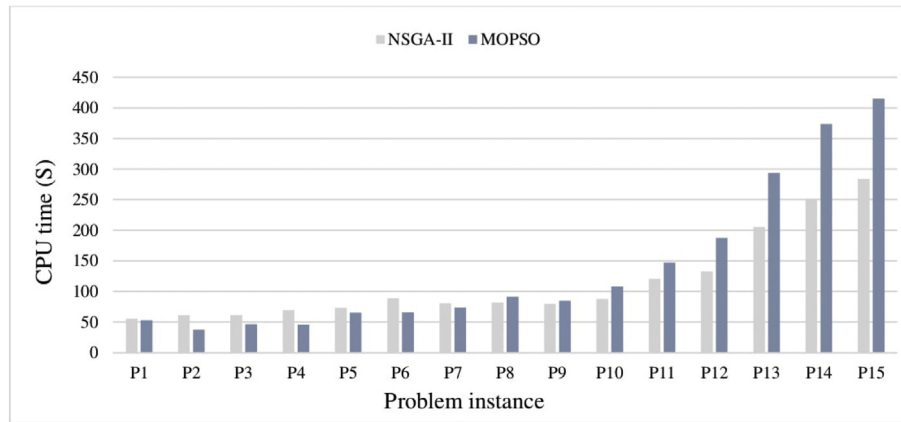


Fig. 9. Comparison of NSGA-II and MOPSO in terms of computational time.

$$Spacing = \sqrt{\frac{1}{N-1} \sum_{i=1}^N (d_i - \bar{d})^2} \quad (38)$$

$$d_i = \min_j \sum_{t=1}^T |f_t^j - f_t^i| \quad \forall i, j \in \{1, \dots, N\} \quad (39)$$

where N is the number of non-dominated solutions, d_i is the minimum distance of Pareto optimal solution i from other solutions, $\bar{d}$ is the mean of d_i for all non-dominated solutions, and the value of objective function $t \in \{1, 2, \dots, T\}$ for Pareto optimal solution i is denoted by f_t^i .

- The diversity metric

This metric calculated as the maximum Euclidean distance between non-dominated solutions. An algorithm with greater value of diversity can cover a larger area of solutions.

The computational results of comparing the solution approaches for solving the problem instances in various types are reported in Figs. 9–12. In addition, details of the obtained results have been presented in Appendix B. Two methods are compared to each other based on each considered criterion using two sample t -test. The comparison results for different comparison metrics are presented in Table 6.

Based on the obtained results, NSGA-II has a better performance than MOPSO in exploring more Pareto optimal solutions

with more uniformity in the solution space, while MOPSO behaves better than NSGA-II in producing diverse non-dominated solutions. Furthermore, the p -value of the test-statistic on the equality of the means of the CPU times associated with each solution approach states that there are no significant differences between their means. This is because that in small scale instances, the CPU times for both algorithms are nearly in the same range while by increasing dimension of the problem, NSGA-II outperforms the second algorithm with respect to the computational time. Therefore, although it seems that the capability of the NSGA-II is superior to that of the MOPSO procedure in handling the large-scale category of the location-routing problem of this study, awareness of its weaknesses (i.e. low diversity of its solutions) can lead to the incorporation of design improvements into its current framework in future studies.

6. Conclusion

This study has developed a new mathematical formulation for the industrial hazardous waste location-routing problems with three objective functions designed to keep a balance and trade-off between economic and environmental considerations. Taking into account some realistic assumptions disregarded in the previous studies such as heterogeneous collection vehicle fleet with different capacity and maximum allowable travelled distance and waste-waste incompatibility restrictions as well as the incorporation of the vehicle routing decisions into the model are the main contributions of this study. Validation of the proposed model has been carried out by solving a small scale problem instance using

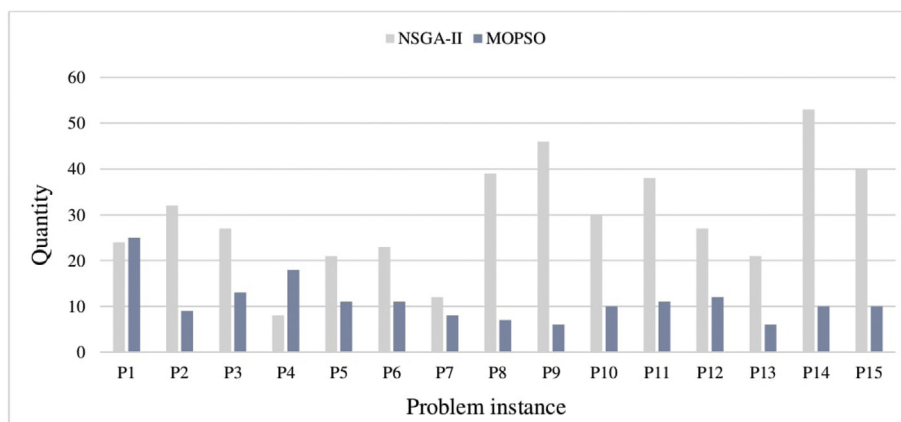


Fig. 10. Comparison of NSGA-II and MOPSO in terms of quantity metric.

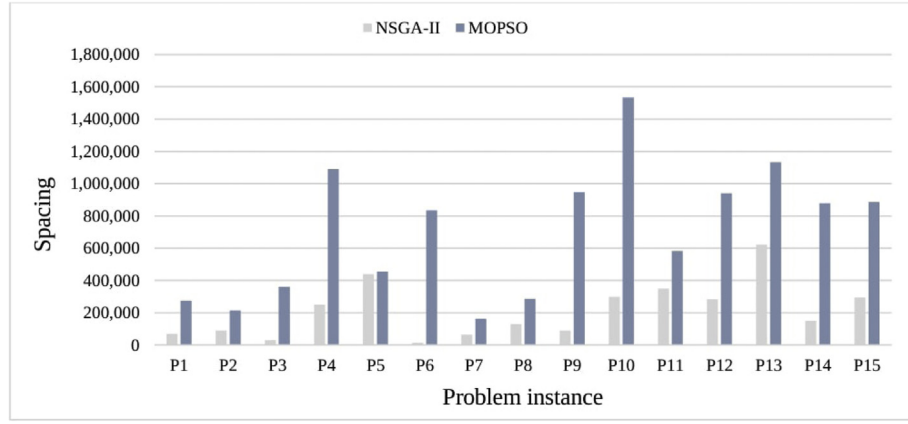


Fig. 11. Comparison of NSGA-II and MOPSO in terms of spacing metric.

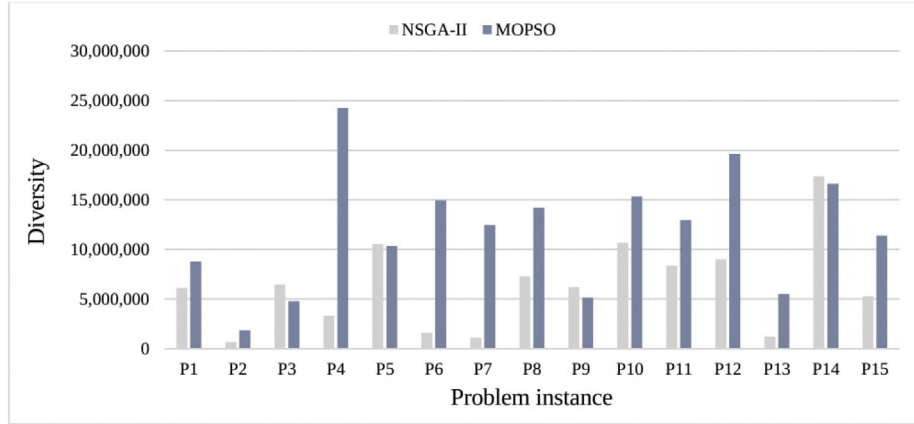


Fig. 12. Comparison of NSGA-II and MOPSO in terms of diversity metric.

Table 6
Comparison of solution approaches using two sample *t*-test.

Criteria	Optimal method	Average results		Two sample <i>t</i> -test	
		NSGA-II	MOPSO	Comparison of means	P-value
Quantity	NSGA-II	29	11	$\mu_{NSGA-II} > \mu_{MOPSO}$	0.000
CPU time (s)	NSGA-II and MOPSO	115.63	139.37	$\mu_{NSGA-II} = \mu_{MOPSO}$	0.528
Spacing	NSGA-II	211000	705000	$\mu_{NSGA-II} < \mu_{MOPSO}$	0.000
Diversity	MOPSO	6340000	11880000	$\mu_{NSGA-II} < \mu_{MOPSO}$	0.009

GAMS software. Applicability of the evolutionary computing methods in this area is another contribution of this study. In this regard, both NSGA-II and MOPSO algorithms were employed to solve the different sets of this NP-hard problem efficiently. Finally, the performance and accuracy of the solution approaches was compared based upon different metrics. The results indicated that according to both of the performance metrics quantity and spacing, NSGA-II had a better performance than the other approach while the diversity metric of MOPSO is better than NSGA-II. Furthermore, for largescale problems, the required CPU time for MOPSO was significantly more than NSGA-II. Therefore, although it is increasingly clear that NSGA-II should be introduced as the better solution approach in handling the location-routing problem of this study, awareness of its weaknesses can lead to the incorporation of design improvements into its current framework in future studies. Next, we will extend this work by considering the inherent uncertainty of the problem parameters such as an amount of waste generated at each region to solve a more realistic problem. Stochastic programming or robust optimization approach can be promising tools for this purpose.

Appendix A. Model linearization

Nonlinearity of the model is resulted from appearance of the products $x_{ijk} l_{ik}$ in the mentioned nonlinear terms in sub section 2.1. A simple method to evade this nonlinearity caused by $x_{ijk} l_{ik}$ is defining auxiliary variable xl_{ijk} and replacing $x_{ijk} l_{ik}$ by it. Also, we must add additional constraints A.1-A.4 to the model.

$$xl_{ijk} \leq M \quad x_{ijk} \quad \forall i \in G, j \in G \cup RuT, k \in K \quad (A.1)$$

$$xl_{ijk} \leq l_{ik} \quad \forall i \in G, j \in G \cup RuT, k \in K \quad (A.2)$$

$$xl_{ijk} \geq l_{ik} - (1 - x_{ijk}) M \quad \forall i \in G, j \in G \cup RuT, k \in K \quad (A.3)$$

$$xl_{ijk} \geq 0, M > \infty \quad \forall i \in G, j \in G \cup RuT, k \in K \quad (A.4)$$

In our problem, we set the value of large number M as $\sum_{w \in W} v_{wk} \delta_w$. By using the same approach, the nonlinearity associated with $xt_{wi} t_{qi}$ is also removable.

Appendix B. Comparison metrics for problem instances

Table B.7

Comparison metrics for problem instances

Problem number	Problem dimension	NSGA-II				MOPSO			Diversity ($\times 10^6$)
		CPU time (s)	Quantity	Spacing ($\times 10^5$)	Diversity ($\times 10^6$)	CPU time	Quantity	Spacing ($\times 10^5$)	
1	(6, 2/2, 2/1, 2/2)	55.33	24	0.7035	6.125	52.90	25	2.771	8.783
2	(8, 3/2, 2/2, 2/2)	61.51	32	0.8969	0.6594	37.45	9	2.148	1.855
3	(10, 3/3, 2/2, 3/2)	61.37	27	0.3021	6.447	46.49	13	3.596	4.774
4	(12, 4/3, 3/2, 4/2)	69.48	8	2.515	3.305	46.02	18	10.92	24.28
5	(15, 4/4, 3/2, 4/3)	73.39	21	4.393	10.53	65.32	11	4.541	10.35
6	(20, 5/3, 3/3, 4/3)	88.88	23	0.1446	1.607	66.12	11	8.346	14.97
7	(20, 5/4, 4/3, 4/4)	81.10	12	0.6428	1.104	73.91	8	1.626	12.45
8	(25, 6/4, 4/3, 5/4)	82.09	39	1.298	7.283	91.14	7	2.874	14.21
9	(25, 7/3, 4/4, 5/4)	79.81	46	0.8924	6.175	84.96	6	9.483	5.14
10	(30, 7/4, 5/3, 6/4)	87.84	30	2.989	10.69	108.48	10	15.34	15.32
11	(40, 8/5, 5/5, 7/4)	120.66	38	3.507	8.341	147.43	11	5.827	12.97
12	(50, 9/7, 6/5, 9/5)	132.80	27	2.836	9.015	187.54	12	9.41	19.62
13	(75, 10/7, 6/6, 9/6)	205.50	21	6.228	1.227	294.01	6	11.33	5.535
14	(90, 12/8, 7/6, 10/7)	251.05	53	1.487	17.4	373.65	10	8.776	16.61
15	(100, 15/9, 8/6, 12/8)	283.67	40	2.948	5.258	415.14	10	8.874	11.42
Average	115.63	29	2.11	6.34	139.37	11	7.05	11.88	

Appendix C. Supplementary data

Supplementary data related to this article can be found at <http://dx.doi.org/10.1016/j.jclepro.2017.09.029>.

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